

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (EC) Examination

November 2013

EC 303/EC 303.01 Audio Video Engineering

Date: 22.11.2013, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Fill in the blanks with appropriate **digits**. [03]

- (1) Threshold of hearing: _____ Pa.
- (2) Threshold of pain: _____ dB.
- (3) Range of audio frequency is _____ Hz to _____ Hz.
- (4) Ribbon microphone should be kept about _____ meter or more away from the source of sound.
- (5) Impedance of the Crystal microphone is about _____ MΩ.
- (6) Nominal absorption coefficient for audience is _____.

(b) Define following terms. [04]

- | | |
|----------------------|---------------|
| (1) SRI | (2) Sone |
| (3) Sabine's Formula | (4) Overtones |
| (5) Intervals | (6) Harmonics |
| (7) Octaves | (8) Timbre |

Q - 2 (a) With the help of neat sketch explain the microphone used in Telephone. [05]

(b) Calculate the directivity of the directive microphone whose output in the best direction is 1mV, when the output of an omni-directional microphone placed under identical conditions with respect to source is only 100μV. [04]

(c) There is a hall of 10m × 8m × 3m having following surfaces: [05]

Absorbent Material	Dimensions	Nos.	Absorption Co-efficient
Open window	2 × 2m	7Nos.	1
Thin Carpet	10 × 5m	1No.	0.25
Galeecha	15 × 10m	1No.	0.58
Curtains	2.5 × 2m	7Nos.	0.15
Audiences		330	0.45

Calculate reverberation time.

OR

Q - 2 (a) Explain the microphone which has highest impedance. [05]

(b) The output of a directive microphone is 0.3mV, when the output of an omni-directional microphone is 3μV. Calculate the directivity of microphone. Also calculate the directivity of an omni-directional microphone. [04]

(c) Calculate the reverberation time for a hall whose length, width and height are 50m, 30m and 5m, respectively, and the average absorption co-efficient is 0.161. [05]

Q - 3 (a) Write Short note on followings: [08]

- (1) 3D Television
- (2) Hypertext, Hypermedia and Multimedia

- (b) Give salient features of acoustical design for an auditorium. [04]
- (c) Compare cone type and horn type loudspeaker. [02]

OR

- Q - 3 (a)** Write Short note on followings: [08]
- (1) Digital Television
 - (2) Advantages and Disadvantages of Multimedia
- (b) Explain Growth and decay of sound in an enclosure and specially mention the reverberation time for nearly dead rooms. [04]
 - (c) What is cross-over network? Explain its necessity. [02]

SECTION – II

- Q - 4 (a)** How is the Illusion of continuity created in television pictures? Why has the frame reception rate been chosen to be 25 and not 24 as in motion pictures? [03]
- (b) What is a ghost image and what causes it to appear on the receiver screen along with the reproduced picture. What causes trailing and leading ghost images? [04]
- Q - 5 (a)** Justify the need for pre- and post equalizing pulses. Why it is necessary to keep their duration equal to the half line period? [05]
- (b) Define Followings. [02]
 - (1) Front Porch
 - (2) Back Porch
 - (c) Draw the block diagram of an RF tuner and explain how incoming signals from different station are translated to common picture IF and sound IF frequencies. [07]

OR

- Q - 5 (a)** Sketch composite video signal waveform for at least three successive lines and indicate (I) Extreme white level (II) Blanking Level (III) Pedestal height (IV) Sync Pulse level. [05]
- (b) What is the function of vertical Pulse train [02]
 - (c) Describe the factor which influences the choice of intermediate frequency in T.V. receiver. [07]
- Q - 6 (a)** Explain the Automatic Gain Control in T.V. receiver [03]
- (b) What do you understand by interlaced scanning? Show that it reduces flicker and conserves bandwidth. [05]
 - (c) Show that a total channel bandwidth of 11.25MHz would be necessary if both sidebands of the modulated signals are fully transmitted. Explain how the channel width is reduced to 7MHz by allowing vestigial side band transmission. [06]

OR

- Q - 6 (a)** With neat sketch diagram explain the Video detector in T.V. receiver [03]
- (b) Describe the meaning of luminance, Hue, Saturation, Brightness, Contrast as applied to colour pictures. [05]
 - (c) Explain why transmission is done at frequencies above 40MHz. Justify that space wave propagation is the only effective for the TV signal radiation above about 40MHz. [06]